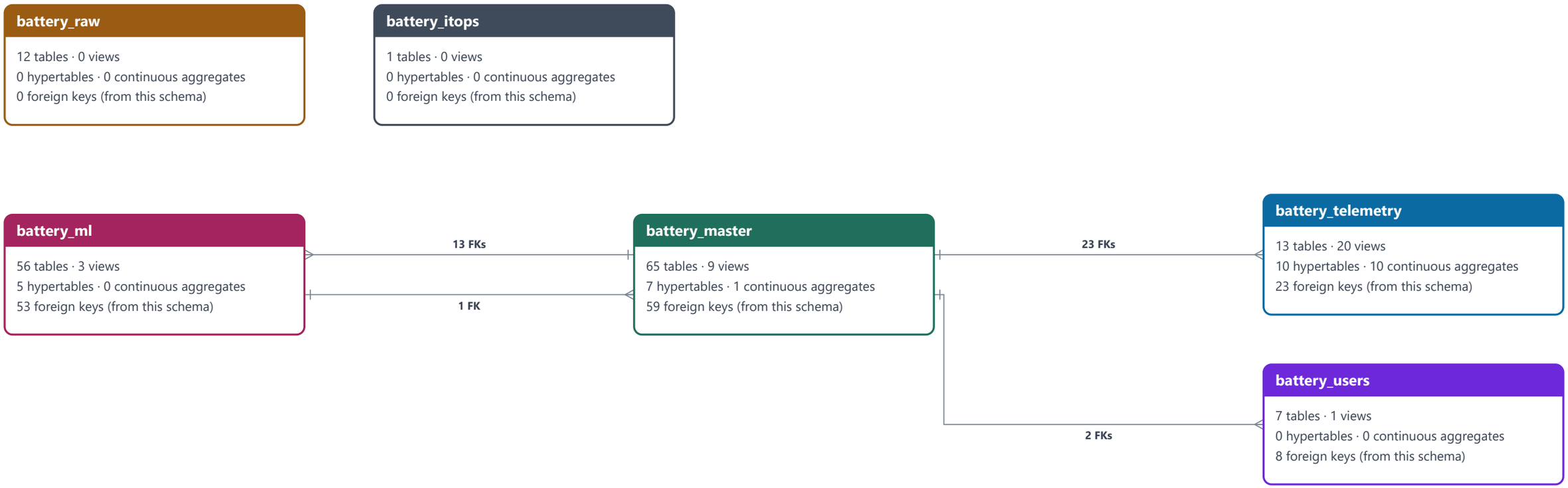
BESS database — entity relationship diagrams

Database **Battery** · generated 2026-10-22 · 154 tables · 33 views · 143 foreign keys



Legend

PK primary key · FK foreign key · PF both · UQ unique • NOT NULL · hypertable TimescaleDB hypertable

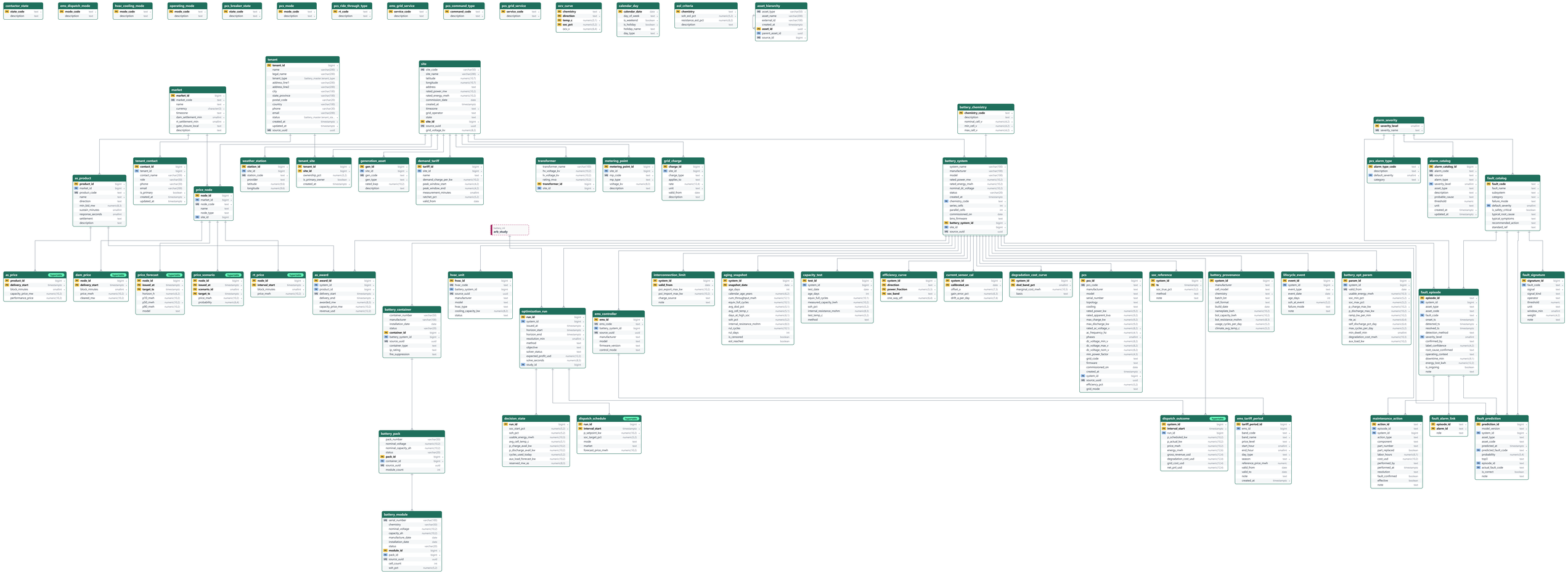
Lines run from the referenced (parent) table, marked with a bar, to the referencing (child) table, marked with a crow's foot. Referenced tables in other schemas appear as dashed boxes. Arrows on this page count foreign keys between schemas.

DDL for every object is in **BESS DESIGN/Database/DDL** (00_create_all.sql runs everything in order).

battery_master

Master and reference data: tenants, sites, battery systems and their assets, catalogues, market and dispatch data

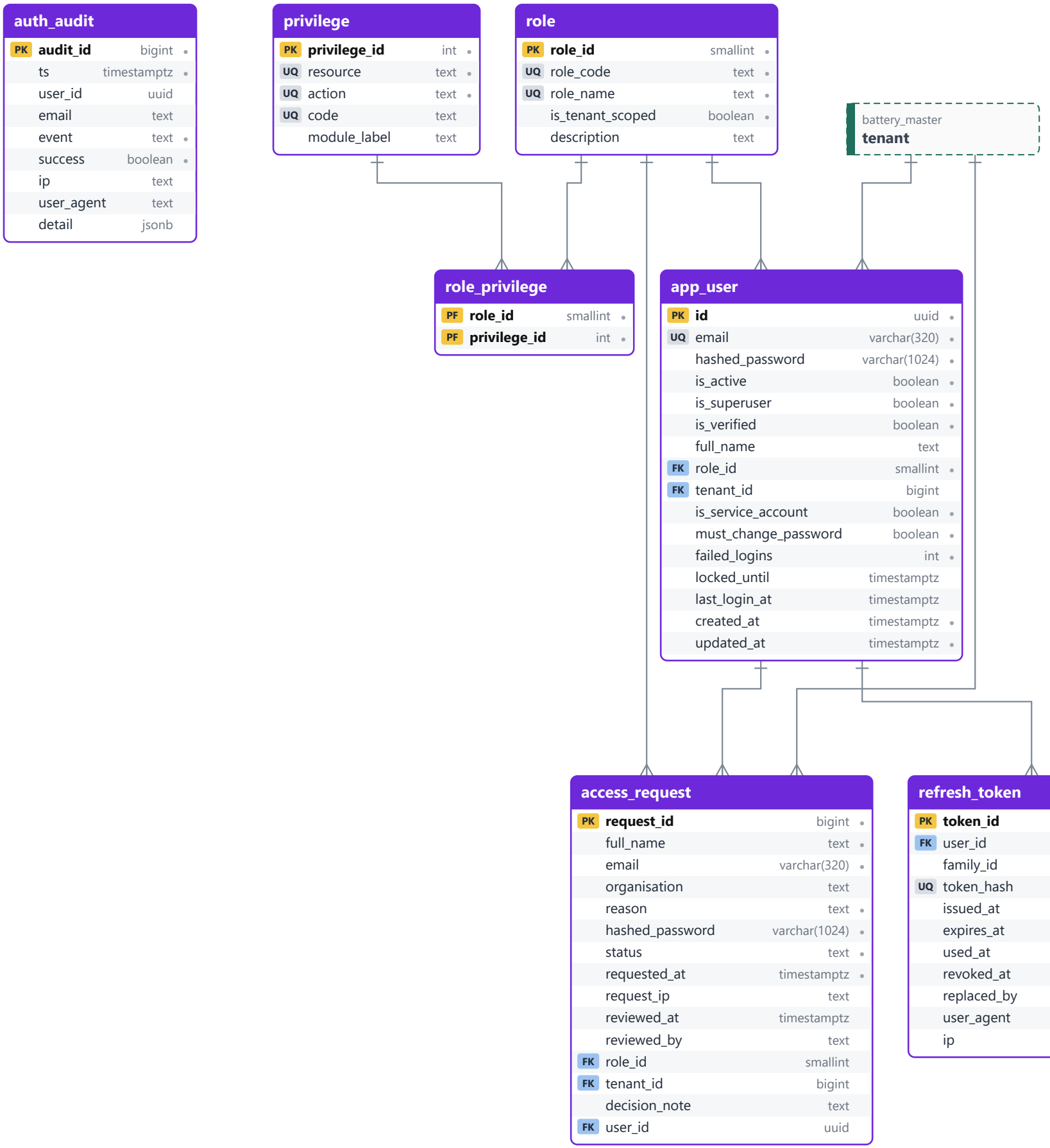
63 tables · 7 hypertables · 59 foreign keys · 9 views (see index) · dashed boxes are tables in other schemas



battery_users

Login, roles, permissions, refresh tokens, access requests and the auth audit trail

7 tables · 0 hypertables · 8 foreign keys · 1 views (see index) · dashed boxes are tables in other schemas



battery_raw

Staging tables and batch logs for the loaders

12 tables · 0 hypertables · 0 foreign keys · 0 views (see index) · dashed boxes are tables in other schemas

bak_20261015_bess_system

	system_id	bigint
	system_code	text

soc_stg

PK	**stg_id**	bigint	•
	system_code	text	
	ts	timestamp	
	soc_true_pct	numeric	
	method	text	
	note	text	
UQ	source_file	text	•
UQ	row_no	int	•
	loaded_at	timestamptz	•

bak_20261015_lifecycle_event

	event_id	bigint
	system_id	bigint
	event_type	text
	event_date	date
	age_days	int
	soh_at_event	numeric(5,2)
	failure_mode	text
	note	text

bak_20261015_rul_backtest

	run_id	bigint
	system_id	bigint
	snapshot_date	date
	method	text
	p10_days	real
	p50_days	real
	p90_days	real
	p50_cycles	real
	beyond	boolean
	actual_days	real
	actual_cycles	real
	known_alive_days	real
	censored	boolean

bak_20261015_rul_forecast

	run_id	bigint
	system_id	bigint
	method	text
	as_of	date
	soh_pct	real
	resistance_ratio	real
	p10_days	real
	p50_days	real
	p90_days	real
	p10_cycles	real
	p50_cycles	real
	p90_cycles	real
	eol_date_p50	date
	trigger	text
	beyond	boolean

bak_20261015_capacity_test

	test_id	bigint
	system_id	bigint
	test_date	date
	age_days	int
	equiv_full_cycles	numeric(10,1)
	measured_capacity_kwh	numeric(10,2)
	soh_pct	numeric(5,2)
	internal_resistance_mohm	numeric(8,3)
	test_temp_c	numeric(5,2)
	method	text

bak_20261015_rul_score

	run_id	bigint
	method	text
	scope	text
	n	int
	mae_days	numeric
	rmse_days	numeric
	bias_days	numeric
	mae_cycles	numeric
	coverage_pct	numeric
	pinball_days	numeric
	c_index	numeric
	censored_violation_pct	numeric
	oracle_mae_days	numeric

bak_20261015_battery_provenance

	system_id	bigint
	manufacturer	text
	cell_model	text
	chemistry	text
	batch_lot	text
	cell_format	text
	build_date	date
	nameplate_kwh	numeric(10,2)
	bol_capacity_kwh	numeric(10,2)
	bol_resistance_mohm	numeric(8,3)
	usage_cycles_per_day	numeric(5,2)
	climate_avg_temp_c	numeric(5,2)

bak_20261015_rul_run

	run_id	bigint
	use_case	text
	run_name	text
	status	text
	is_default	boolean
	system_ids	bigint[]
	methods	text[]
	params	jsonb
	n_units	int
	n_rows	int
	metrics	jsonb
	results	jsonb
	warnings	jsonb
	error_message	text
	artifact_path	text
	started_at	timestamptz
	finished_at	timestamptz
	duration_ms	int

soc_estimation_data

PK	**stg_id**	bigint	•
	system_name	text	•
	ts	timestamp	
	true_current_a	numeric	
	measured_current_a	numeric	
	current_sensor_offset_a	numeric	
	soc_true_pct	numeric	
	soc_coulomb_pct	numeric	
	soc_bms_pct	numeric	
	usable_capacity_ah	numeric	
	coulombic_efficiency	numeric	
	at_rest	boolean	
	rest_duration_min	numeric	
	ocv_rested_v	numeric	
UQ	source_file	text	•
UQ	row_no	int	•
	loaded_at	timestamptz	•

bak_20261015_aging_snapshot

	system_id	bigint
	snapshot_date	date
	age_days	int
	calendar_age_years	numeric(6,2)
	cum_throughput_mwh	numeric(12,1)
	equiv_full_cycles	numeric(10,1)
	avg_dod_pct	numeric(5,1)
	avg_cell_temp_c	numeric(5,1)
	days_at_high_soc	numeric(6,1)
	soh_pct	numeric(5,2)
	internal_resistance_mohm	numeric(8,3)
	rul_cycles	numeric(10,1)
	rul_days	int
	is_censored	boolean
	eol_reached	boolean

batchlog

PK	**batch_id**	bigint	•
	run_id	uuid	•
	job_name	text	•
	file_name	text	•
	file_time	timestamptz	
	snapshot_time	timestamptz	
	status	text	•
	started_at	timestamptz	•
	finished_at	timestamptz	•
	duration_ms	int	•
	feeds_total	int	•
	feeds_loaded	int	•
	feeds_failed	int	•
	rows_total	int	•
	rows_inserted	int	•
	rows_updated	int	•
	rows_skipped	int	•
	feed_results	jsonb	•
	warnings	jsonb	•
	id_changes	jsonb	•
	error_message	text	
	archive_path	text	
	api_url	text	
	host_name	text	
	report	jsonb	
	created_at	timestamptz	•

battery_telemetry

Time-series telemetry (TimescaleDB hypertables), alarms, dispatch events and their rollups

13 tables · 10 hypertables · 23 foreign keys · 20 views (see index) · dashed boxes are tables in other schemas

weather_station

PK	**station_id**	bigint
UQ	station_code	text

metering_point

PK	**metering_point_id**	bigint
UQ	mp_code	text

calendar_day

PK	**calendar_date**	date
	day_type	text
	is_weekend	boolean
	is_holiday	boolean

battery_master
battery_module

module_telemetry — hypertable

PF	**module_id**	bigint
PK	**ts**	timestamptz
	cell_v_avg_v	numeric(6,4)
	cell_v_min_v	numeric(6,4)
	cell_v_max_v	numeric(6,4)
	cell_temp_max_c	numeric(6,2)
	cell_temp_min_c	numeric(6,2)
	balancing_active	boolean

battery_master
metering_point

demand_telemetry — hypertable

PF	**metering_point_id**	bigint
PK	**ts**	timestamptz
	gross_load_kw	numeric(12,2)
	behind_meter_gen_kw	numeric(12,2)
	net_load_kw	numeric(12,2)
	load_reactive_kvar	numeric(12,2)

battery_master
weather_station

weather_actual — hypertable

PF	**station_id**	bigint
PK	**ts**	timestamptz
	temp_c	numeric(6,2)
	ghi_wm2	numeric(8,1)
	humidity_pct	numeric(5,1)
	wind_ms	numeric(5,1)
	cloud_fraction	numeric(4,3)

weather_forecast — hypertable

PF	**station_id**	bigint
PK	**issued_at**	timestamptz
PK	**valid_at**	timestamptz
	temp_c	numeric(6,2)
	ghi_wm2	numeric(8,1)

battery_master
hvac_cooling_mode

battery_master
hvac_unit

hvac_telemetry — hypertable

PF	**hvac_id**	bigint
PK	**ts**	timestamptz
FK	cooling_mode	text
	ambient_temp_c	numeric(6,2)
	coolant_setpoint_c	numeric(5,2)
	coolant_supply_temp_c	numeric(6,2)
	coolant_return_temp_c	numeric(6,2)
	coolant_delta_t_c	numeric(6,2)
	coolant_flow_lpm	numeric(8,2)
	compressor_speed_pct	numeric(5,2)
	condenser_fan_pct	numeric(5,2)
	refrig_suction_bar	numeric(7,2)
	refrig_discharge_bar	numeric(7,2)
	battery_heat_load_kw	numeric(9,2)
	cooling_capacity_kw	numeric(9,2)
	heat_rejected_kw	numeric(9,2)
	hvac_power_kw	numeric(9,2)
	cop	numeric(5,2)
	enclosure_temp_c	numeric(6,2)
	enclosure_humidity_pct	numeric(5,2)
	filter_dp_pa	numeric(8,1)
	cum_energy_kwh	numeric(13,2)
	compressor_runtime_h	numeric(10,2)
	units_running	smallint
	alarm_level	smallint
	alarm_flags	text

battery_master
pcs_breaker_state

battery_master
pcs_grid_service

battery_master
pcs_mode

battery_master
pcs

pcs_telemetry — hypertable

PF	**pcs_id**	bigint
PK	**ts**	timestamptz
FK	mode_code	text
FK	grid_service	text
	ac_active_power_kw	numeric(9,2)
	ac_reactive_power_kvar	numeric(9,2)
	ac_apparent_power_kva	numeric(9,2)
	power_factor	numeric(5,3)
	ac_frequency_hz	numeric(6,3)
	ac_voltage_ab_v	numeric(8,2)
	ac_voltage_bc_v	numeric(8,2)
	ac_voltage_ca_v	numeric(8,2)
	ac_current_a_a	numeric(9,2)
	ac_current_b_a	numeric(9,2)
	ac_current_c_a	numeric(9,2)
	thd_voltage_pct	numeric(5,2)
	thd_current_pct	numeric(5,2)
	dc_voltage_v	numeric(8,2)
	dc_current_a	numeric(9,2)
	dc_power_kw	numeric(9,2)
	efficiency_pct	numeric(5,2)
	igbt_temp_c	numeric(6,2)
	heatsink_temp_c	numeric(6,2)
	inlet_coolant_temp_c	numeric(6,2)
	ambient_temp_c	numeric(6,2)
	avail_charge_kw	numeric(9,2)
	avail_discharge_kw	numeric(9,2)
	ac_energy_import_kwh	numeric(13,2)
	ac_energy_export_kwh	numeric(13,2)
FK	breaker_state	text
	alarm_level	smallint

battery_master
operating_mode

battery_master
contactor_state

battery_master
battery_system

battery_master
alarm_severity

battery_master
alarm_catalog

battery_master
ems_dispatch_mode

battery_master
ems_grid_service

battery_master
ems_controller

system_telemetry — hypertable

PK	**ts**	timestamptz
FK	mode_code	text
	soc_bms_pct	numeric(5,2)
	soh_pct	numeric(5,2)
	pack_voltage_v	numeric(8,2)
	pack_current_a	numeric(9,2)
	power_kw	numeric(9,2)
	c_rate	numeric(6,3)
	cell_v_avg_v	numeric(6,4)
	cell_v_min_v	numeric(6,4)
	cell_v_max_v	numeric(6,4)
	cell_v_delta_mv	numeric(7,2)
	cell_temp_avg_c	numeric(6,2)
	cell_temp_min_c	numeric(6,2)
	cell_temp_max_c	numeric(6,2)
	coolant_temp_c	numeric(6,2)
	ambient_temp_c	numeric(6,2)
	insulation_res_kohm	numeric(9,1)
	avail_charge_kw	numeric(9,2)
	avail_discharge_kw	numeric(9,2)
	cum_charge_kwh	numeric(12,2)
	cum_discharge_kwh	numeric(12,2)
	equiv_full_cycles	numeric(10,3)
FK	contactor_state	text
	alarm_level	smallint
PF	**system_id**	bigint
	measured_current_a	numeric(9,2)
	current_sensor_offset_a	numeric(7,3)
	soc_coulomb_pct	numeric(5,2)
	soc_battery_telemetry_pct	numeric(5,2)
	usable_capacity_ah	numeric(9,2)
	coulombic_efficiency	numeric(6,4)
	at_rest	boolean
	rest_duration_min	numeric(7,1)
	ocv_rested_v	numeric(6,4)

alarm_event — hypertable

PK	**time**	timestamptz
PK	**alarm_id**	text
FK	alarm_catalog_id	bigint
FK	severity_level	smallint
	asset_type	text
	asset_code	text
	linked_asset_type	text
	linked_asset_id	bigint
	asset_match	text
FK	battery_system_id	bigint
	end_time	timestamptz
	trigger_value	numeric
	threshold_value	numeric
	unit	text
	acknowledged_by	text
	acknowledged_at	timestamptz
	created_at	timestamptz
	updated_at	timestamptz

dispatch_event — hypertable

PK	**time**	timestamptz
FK	battery_system_id	bigint
PK	**dispatch_id**	text
FK	ems_id	bigint
	command_code	text
FK	dispatch_mode	text
FK	grid_service	text
	source	text
	issued_at	timestamptz
	effective_to	timestamptz
	p_setpoint_kw	numeric
	target_soc_pct	numeric
	soc_start_pct	numeric
	soc_end_pct	numeric
	ramp_rate_kw_s	numeric
	import_kwh	numeric
	export_kwh	numeric
	avg_price_mwh	numeric
	revenue_usd	numeric
	accepted	boolean
	status	text
	note	text
	created_at	timestamptz
	updated_at	timestamptz

ems_telemetry — hypertable

PF	**ems_id**	bigint
PK	**ts**	timestamptz
FK	dispatch_mode	text
FK	grid_service	text
	market_price_mwh	numeric(10,2)
	p_setpoint_kw	numeric(10,2)
	poi_active_power_kw	numeric(10,2)
	poi_reactive_power_kvar	numeric(10,2)
	tracking_error_kw	numeric(10,2)
	grid_frequency_hz	numeric(6,3)
	poi_voltage_kv	numeric(8,3)
	agc_signal	numeric(6,3)
	plant_soc_pct	numeric(5,2)
	target_soc_pct	numeric(5,2)
	avail_charge_kw	numeric(10,2)
	avail_discharge_kw	numeric(10,2)
	avail_charge_energy_kwh	numeric(12,2)
	avail_discharge_energy_kwh	numeric(12,2)
	aux_load_kw	numeric(9,2)
	aux_load_pct	numeric(6,2)
	cum_import_kwh	numeric(14,2)
	cum_export_kwh	numeric(14,2)
	revenue_cum_usd	numeric(14,2)
	units_online	smallint
	plant_available	smallint
	alarm_level	smallint

battery_ml

ML runs, scores, predictions, KPIs and optimiser results

56 tables · 5 hypertables · 53 foreign keys · 3 views (see index) · dashed boxes are tables in other schemas

hvac_cop_run

PK **run_id**	bigint
run_name	text
status	text
hvac_ids	bigint[]
date_from	date
date_to	date
params	jsonb
models	jsonb
summary	jsonb
warnings	jsonb
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

pcs_kpi_run

PK **run_id**	bigint
status	text
triggered_by	text
pcs_ids	bigint[]
start_date	date
end_date	date
config_file	text
config_version	int
config_hash	text
kpi_codes	text[]
systems_done	int
days	int
rows_deleted	int
rows_written	int
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

ems_kpi_run

PK **run_id**	bigint
status	text
triggered_by	text
ems_ids	bigint[]
start_date	date
end_date	date
config_file	text
config_version	int
config_hash	text
kpi_codes	text[]
systems_done	int
days	int
rows_deleted	int
rows_written	int
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

battery_master
pcs

battery_master
ems_controller

hvac_cop_point hypertable

PF **run_id**	bigint
PK **hvac_id**	bigint
PK **ts**	timestamptz
mode	text
steady	boolean
is_train	boolean
cop_actual	numeric(8,4)
cop_logged	numeric(8,4)
cop_ratio	numeric(8,4)
cop_loop	numeric(8,4)
cop_map	numeric(8,4)
cop_grey	numeric(8,4)
cop_hybrid	numeric(8,4)
cop_expected	numeric(8,4)
cop_residual	numeric(8,4)
eta_2nd_law	numeric(8,5)
cop_carnot	numeric(7,4)
lift_k	numeric(8,3)
plr	numeric(7,4)
cap_actual_kw	numeric(10,3)
cap_expected_kw	numeric(10,3)
q_dt_kw	numeric(10,3)
power_kw	numeric(10,3)
power_expected_kw	numeric(10,3)
rej_closure_kw	numeric(10,3)
t_amb_c	numeric(7,3)
t_supply_c	numeric(7,3)

pcs_kpi

PF **pcs_id**	bigint
PK **kpi_date**	date
PK **kpi_code**	text
kpi_name	text
unit	text
value	numeric
sample_count	int
config_version	int
config_hash	text
FK run_id	bigint
calculated_at	timestamptz

ems_kpi

PF **ems_id**	bigint
PK **kpi_date**	date
PK **kpi_code**	text
kpi_name	text
unit	text
value	numeric
sample_count	int
config_version	int
config_hash	text
FK run_id	bigint
calculated_at	timestamptz

model_run

PK **run_id**	bigint
run_name	text
model_type	text
status	text
params	jsonb
target	text
grain	text
date_from	date
date_to	date
feature_groups	text[]
features	text[]
module_rows_only	boolean
include_snapshots	boolean
split_method	text
test_fraction	numeric
random_state	int
n_rows	int
n_train	int
n_test	int
train_from	timestamptz
train_to	timestamptz
test_from	timestamptz
test_to	timestamptz
target_min	numeric
target_max	numeric
target_std	numeric
r2	numeric
mae	numeric
rmse	numeric
mape_pct	numeric
max_error	numeric
median_ae	numeric
explained_variance	numeric
train_r2	numeric
train_mae	numeric
train_rmse	numeric
iterations	int
best_iteration	int
feature_importance	jsonb
feature_ranges	jsonb
predictions	jsonb
warnings	jsonb
error_message	text
artifact_path	text
is_active	boolean
started_at	timestamptz
finished_at	timestamptz
duration_ms	int
min_importance_pct	numeric
importance_method	text
candidate_features	text[]
selection_importance	jsonb
use_case	text
scope	text
site_id	bigint
battery_system_id	bigint
training_mode	text
FK parent_run_id	bigint
systems	bigint[]
holdout_system_id	bigint
metrics_by_system	jsonb
holdout_metrics	jsonb

soc_forecast_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
ems_ids	bigint[]
date_from	date
date_to	date
horizon_h	numeric
test_fraction	numeric
origin_step_min	int
params	jsonb
n_train_rows	int
n_test_origins	int
best_method	text
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

soc_forecast_point

PF **run_id**	bigint
PK **ems_id**	bigint
PK **origin_ts**	timestamptz
PK **step**	smallint
PK **method**	text
p10	real
p50	real
p90	real
actual	real
plan_kw	real

soc_forecast_score

PF **run_id**	bigint
PK **method**	text
n	int
mae	numeric
rmse	numeric
mae_1h	numeric
mae_4h	numeric
mae_end	numeric
max_error	numeric
coverage_pct	numeric
width	numeric
skill_pct	numeric
drawdown_mae	numeric
breach_pct	numeric

thermal_forecast_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
system_ids	bigint[]
date_from	date
date_to	date
targets	text[]
horizons_min	integer[]
test_fraction	numeric
params	jsonb
n_train_rows	int
n_test_rows	int
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

thermal_forecast_point

PF **run_id**	bigint
PK **target**	text
PK **system_id**	bigint
PK **origin_ts**	timestamptz
PK **horizon_min**	smallint
PK **method**	text
p10	real
p50	real
p90	real
actual	real
now_c	real

thermal_forecast_score

PF **run_id**	bigint
PK **target**	text
PK **horizon_min**	int
PK **method**	text
n	int
mae	numeric
rmse	numeric
pinball	numeric
coverage_pct	numeric
threshold_c	numeric
events	int
recall_p50	numeric
precision_p50	numeric
recall_p90	numeric
precision_p90	numeric

price_forecast_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
ems_ids	bigint[]
date_from	date
date_to	date
resolution	text
horizon_hours	int
test_fraction	numeric
origin_step_h	int
feature_groups	text[]
models	text[]
params	jsonb
n_hours	int
n_train_rows	int
n_test_origins	int
n_spikes	int
best_model	text
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

price_forecast_point

PF **run_id**	bigint
PK **ems_id**	bigint
PK **origin_ts**	timestamptz
target_ts	timestamptz
PK **horizon_h**	smallint
PK **model**	text
p10	real
p50	real
p90	real
mean	real
spike_prob	real
actual	real

price_forecast_score

PF **run_id**	bigint
PK **model**	text
n	int
mae	numeric
rmse	numeric
smape_pct	numeric
pinball	numeric
coverage_pct	numeric
width_mwh	numeric
skill_mae_pct	numeric
peak_mae	numeric
spike_recall	numeric
spike_precision	numeric
spike_brier	numeric
spread_capture_pct	numeric

aux_forecast_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
ems_ids	bigint[]
date_from	date
date_to	date
horizon_hours	int
test_fraction	numeric
params	jsonb
n_train_rows	int
n_test_rows	int
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

aux_forecast_daily

PF **run_id**	bigint
PK **ems_id**	bigint
PK **day**	date
PK **method**	text
actual_kwh	numeric
p10_kwh	numeric
p50_kwh	numeric
p90_kwh	numeric

aux_forecast_point

PF **run_id**	bigint
PK **ems_id**	bigint
PK **origin_ts**	timestamptz
PK **horizon_min**	smallint
PK **method**	text
p10	real
p50	real
p90	real
actual	real

aux_forecast_score

PF **run_id**	bigint
PK **ems_id**	bigint
PK **horizon_bucket**	text
PK **method**	text
n	int
mae	numeric
rmse	numeric
pinball	numeric
coverage_pct	numeric
bias	numeric

heatload_run

PK **run_id**	bigint
run_name	text
status	text
system_ids	bigint[]
date_from	date
date_to	date
params	jsonb
models	jsonb
summary	jsonb
warnings	jsonb
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

heatload_forecast_point hypertable

PF **run_id**	bigint
PK **system_id**	bigint
PK **issued_at**	timestamptz
PK **horizon_min**	int
PK **valid_at**	timestamptz
actual_kw	numeric(10,4)
persistence_kw	numeric(10,4)
physics_kw	numeric(10,4)
p10_kw	numeric(10,4)
p50_kw	numeric(10,4)
p90_kw	numeric(10,4)
plan_kw	numeric(10,3)

heatload_ledger hypertable

PF **run_id**	bigint
PK **system_id**	bigint
PK **ts**	timestamptz
direction	text
is_train	boolean
q_batt_irr_kw	numeric(10,4)
q_batt_rev_kw	numeric(10,4)
q_pcs_kw	numeric(10,4)
q_aux_kw	numeric(10,4)
q_solar_kw	numeric(10,4)
q_env_kw	numeric(10,4)
q_storage_kw	numeric(10,4)
q_total_kw	numeric(10,4)
q_model_kw	numeric(10,4)
measured_kw	numeric(10,4)
heat_load_sensor_kw	numeric(10,4)
cooling_kw	numeric(10,4)
hvac_kw	numeric(10,4)
residual_kw	numeric(10,4)
t_amb_c	numeric(7,3)
t_enc_c	numeric(7,3)
t_cell_max_c	numeric(7,3)
dt_cell_c	numeric(7,3)
ghi_wm2	numeric(8,2)
ghi_source	text

model_run_curve

PF **run_id**	bigint
PK **iteration**	int
train_rmse	numeric
valid_rmse	numeric
train_mae	numeric
valid_mae	numeric

efficiency_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
pcs_ids	bigint[]
date_from	date
date_to	date
split_method	text
test_fraction	numeric
min_load_frac	numeric
feature_groups	text[]
features	text[]
booster	text
params	jsonb
n_points	int
n_train	int
n_test	int
n_cycles	int
best_model	text
metrics	jsonb
results	jsonb
predictions	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

efficiency_cycle

PF **run_id**	bigint
PK **cycle_no**	int
pcs_id	bigint
battery_system_id	bigint
start_ts	timestamptz
end_ts	timestamptz
data_set	text
closed	boolean
complete	boolean
e_dc_in_kwh	numeric
e_dc_out_kwh	numeric
e_ac_in_kwh	numeric
e_ac_out_kwh	numeric
e_poi_in_kwh	numeric
e_poi_out_kwh	numeric
e_aux_kwh	numeric
e_hvac_kwh	numeric
soc_start_pct	numeric
soc_end_pct	numeric
depth_pct	numeric
rte_dc_pct	numeric
rte_ac_pct	numeric
rte_sys_pct	numeric
rte_dc_pred_pct	numeric
rte_ac_comp_pct	numeric
rte_sys_comp_pct	numeric
eta_chg_meas_pct	numeric
eta_dis_meas_pct	numeric
eta_chg_pred_pct	numeric
eta_dis_pred_pct	numeric
detail	jsonb

efficiency_score

PF **run_id**	bigint
PK **model**	text
PK **dataset**	text
n	int
r2	numeric
mae	numeric
rmse	numeric
mape_pct	numeric
max_error	numeric
median_ae	numeric
loss_mae_kw	numeric
bias_pp	numeric

rul_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
system_ids	bigint[]
methods	text[]
params	jsonb
n_units	int
n_rows	int
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

rul_backtest

PF **run_id**	bigint
PK **system_id**	bigint
PK **snapshot_date**	date
PK **method**	text
p10_days	real
p50_days	real
p90_days	real
p50_cycles	real
beyond	boolean
actual_days	real
actual_cycles	real
known_alive_days	real
censored	boolean

rul_forecast

PF **run_id**	bigint
PK **system_id**	bigint
PK **method**	text
as_of	date
soh_pct	real
resistance_ratio	real
p10_days	real
p50_days	real
p90_days	real
p10_cycles	real
p50_cycles	real
p90_cycles	real
eol_date_p50	date
trigger	text
beyond	boolean

rul_score

PF **run_id**	bigint
PK **method**	text
PK **scope**	text
n	int
mae_days	numeric
rmse_days	numeric
bias_days	numeric
mae_cycles	numeric
coverage_pct	numeric
pinball_days	numeric
c_index	numeric
censored_violation_pct	numeric
oracle_mae_days	numeric

loss_run

PK **run_id**	bigint
run_name	text
status	text
system_ids	bigint[]
date_from	date
date_to	date
params	jsonb
models	jsonb
summary	jsonb
warnings	jsonb
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

loss_daily

PF **run_id**	bigint
PK **system_id**	bigint
PK **day**	date
intervals	int
is_baseline	boolean
e_in_poi_kwh	numeric(14,3)
e_out_poi_kwh	numeric(14,3)
e_in_ac_kwh	numeric(14,3)
e_out_ac_kwh	numeric(14,3)
e_in_term_kwh	numeric(14,3)
e_out_term_kwh	numeric(14,3)
stored_kwh	numeric(14,3)
throughput_kwh	numeric(14,3)
utilization_pct	numeric(8,3)
loss_batt_kwh	numeric(12,3)
loss_dc_kwh	numeric(12,3)
loss_pcs_kwh	numeric(12,3)
loss_xfmr_kwh	numeric(12,3)
aux_hvac_kwh	numeric(12,3)
aux_ctrl_kwh	numeric(12,3)
exp_batt_kwh	numeric(12,3)
exp_pcs_kwh	numeric(12,3)
exp_xfmr_kwh	numeric(12,3)
exp_aux_kwh	numeric(12,3)
exp_standby_kwh	numeric(12,3)
fixed_kwh	numeric(12,3)
variable_kwh	numeric(12,3)
rte_term	numeric(8,5)
rte_dc	numeric(8,5)
rte_ac	numeric(8,5)
rte_poi_excl_aux	numeric(8,5)
rte_plant	numeric(8,5)
rte_expected	numeric(8,5)
performance_ratio	numeric(8,5)
guarantee_gap_pts	numeric(8,4)
closure_kwh	numeric(12,4)
closure_pcs_kwh	numeric(12,4)
avg_price_mwh	numeric(10,3)
loss_usd	numeric(12,4)
excess_usd	numeric(12,4)
extra	jsonb

loss_ledger hypertable

PF **run_id**	bigint
PK **system_id**	bigint
PK **ts**	timestamptz
direction	text
p_poi_kw	numeric(10,3)
p_ac_kw	numeric(10,3)
p_dc_kw	numeric(10,3)
p_term_kw	numeric(10,3)
p_chem_kw	numeric(10,3)
load_frac	numeric(8,5)
loss_batt_kwh	numeric(10,5)
loss_dc_kwh	numeric(10,5)
loss_pcs_kwh	numeric(10,5)
loss_xfmr_kwh	numeric(10,5)
aux_hvac_kwh	numeric(10,5)
aux_ctrl_kwh	numeric(10,5)
standby_kwh	numeric(10,5)
exp_batt_kwh	numeric(10,5)
exp_pcs_kwh	numeric(10,5)
exp_xfmr_kwh	numeric(10,5)
exp_aux_kwh	numeric(10,5)
eta_pcs	numeric(8,5)
eta_batt	numeric(8,5)
closure_kwh	numeric(10,5)
price_mwh	numeric(10,3)

soc_estimation_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
system_ids	bigint[]
date_from	date
date_to	date
test_fraction	numeric
label_source	text
params	jsonb
n_train_rows	int
n_test_rows	int
n_test_anchors	int
best_method	text
primary_scope	text
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

soc_estimation_point

PF **run_id**	bigint
PK **system_id**	bigint
PK **ts**	timestamptz
is_test	boolean
truth	real
anchor	real
anchor_method	text
coulomb	real
ocv_lookup	real
bms	real
ekf	real
ekf_p10	real
ekf_p90	real
gbm	real
gbm_p10	real
gbm_p90	real
hybrid	real
hybrid_p10	real
hybrid_p90	real
ekf_bias	real
ekf_gain	real
recal_kind	text
at_rest	boolean

soc_estimation_score

PF **run_id**	bigint
PK **method**	text
PK **scope**	text
n	int
rmse	numeric
mae	numeric
bias	numeric
max_error	numeric
p95_abs	numeric
coverage_pct	numeric
width	numeric
skill_pct	numeric

soc_estimation_state

PF **run_id**	bigint
PK **system_id**	bigint
state_ts	timestamptz
state	jsonb
updated_at	timestamptz

netload_forecast_run

PK **run_id**	bigint
use_case	text
run_name	text
status	text
is_default	boolean
mp_ids	bigint[]
date_from	date
date_to	date
horizon_hours	int
test_fraction	numeric
origin_step_h	int
feature_groups	text[]
weather_source	text
peak_threshold_kw	numeric
models	text[]
params	jsonb
n_readings	int
n_train_rows	int
n_test_origins	int
n_test_days	int
best_model	text
metrics	jsonb
results	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

netload_forecast_peak

PF **run_id**	bigint
PK **mp_id**	bigint
PK **origin_ts**	timestamptz
PK **target_date**	date
PK **model**	text
lead_h	real
p10	real
p50	real
p90	real
peak_ts	timestamptz
prob_exceed	real
actual	real
actual_ts	timestamptz
threshold_kw	real

netload_forecast_point

PF **run_id**	bigint
PK **mp_id**	bigint
PK **origin_ts**	timestamptz
target_ts	timestamptz
PK **step**	smallint
PK **model**	text
p10	real
p50	real
p90	real
actual	real

netload_forecast_score

PK **run_id**	bigint
PK **model**	text
n	int
mae	numeric
rmse	numeric
mape_pct	numeric
bias	numeric
pinball	numeric
coverage_pct	numeric
width_kw	numeric
skill_mae_pct	numeric
window_mae	numeric
peak_mae_kw	numeric
peak_bias_kw	numeric
peak_time_mae_min	numeric
exceed_brier	numeric
exceed_auc	numeric
exceed_recall	numeric
exceed_precision	numeric

fault_diag_run

PK **run_id**	bigint
run_name	text
status	text
is_default	boolean
date_from	date
date_to	date
params	jsonb
summary	jsonb
metrics	jsonb
warnings	jsonb
artifact_path	text
published_at	timestamptz
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

battery_master
fault_catalog

battery_master
fault_episode

battery_master
fault_prediction

fault_diag_example

PF **run_id**	bigint
PK **example_key**	text
group_key	text
kind	text
episode_id	text
system_id	bigint
window_start	timestamptz
window_end	timestamptz
label	text
predicted	text
probability	numeric(6,4)
correct	boolean
fold	smallint
detail	jsonb

fault_diag_scan_point

PF **run_id**	bigint
PK **system_id**	bigint
PK **ts**	timestamptz
predicted	text
probability	numeric(6,4)
p_none	numeric(6,4)
top3	jsonb
alarms	text[]

fault_diag_candidate

PK **candidate_id**	bigint
FK run_id	bigint
UQ system_id	bigint
first_ts	timestamptz
UQ peak_ts	timestamptz
last_ts	timestamptz
FK fault_code	text
probability	numeric(6,4)
n_windows	int
top3	jsonb
contributions	jsonb
evidence	jsonb
alarms	text[]
asset_code	text
FK episode_id	text
FK actual_fault_code	text
FK prediction_id	bigint
status	text
created_at	timestamptz

bms_kpi_run

PK **run_id**	bigint
status	text
triggered_by	text
system_ids	bigint[]
start_date	date
end_date	date
config_file	text
config_version	int
config_hash	text
kpi_codes	text[]
systems_done	int
days	int
rows_deleted	int
rows_written	int
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

battery_master
battery_system

battery_master
price_node

bms_kpi

PF **system_id**	bigint
PK **kpi_date**	date
PK **kpi_code**	text
kpi_name	text
unit	text
value	numeric
sample_count	int
config_version	int
config_hash	text
FK run_id	bigint
calculated_at	timestamptz

anomaly_run

PK **run_id**	bigint
use_case	text
run_kind	text
FK parent_run_id	bigint
run_name	text
status	text
FK battery_system_id	bigint
system_name	text
date_from	date
date_to	date
grain	text
feature_groups	text[]
features	text[]
params	jsonb
n_rows	int
n_modules	int
n_packs	int
if_anomalies	int
db_noise	int
db_clusters	int
eps_used	numeric
silhouette	numeric
n_weak	int
n_suspect	int
feature_stats	jsonb
warnings	jsonb
error_message	text
artifact_path	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

mv_anomaly_run

PK **run_id**	bigint
run_name	text
run_kind	text
FK parent_run_id	bigint
status	text
is_default	boolean
FK system_id	bigint
date_from	date
date_to	date
eval_from	timestamptz
params	jsonb
features	jsonb
thresholds	jsonb
metrics	jsonb
summary	jsonb
warnings	jsonb
artifact_path	text
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

arb_study

PK **study_id**	bigint
study_name	text
status	text
FK system_id	bigint
FK node_id	bigint
date_from	date
date_to	date
mode	text
params	jsonb
checks	jsonb
warnings	jsonb
error_message	text
started_at	timestamptz
finished_at	timestamptz
duration_ms	int

battery_master
optimization_run

anomaly_point

PF **run_id**	bigint
PK **module_id**	bigint
PK **bucket**	timestamptz
features	jsonb
if_score	numeric
if_flag	boolean
db_label	int

anomaly_module_result

PF **run_id**	bigint
PK **module_id**	bigint
serial_number	text
pack_id	bigint
pack_number	text
position	int
verdict	text
combined_score	numeric
rank	int
n_points	int
if_score_mean	numeric
if_anomaly_share	numeric
db_noise_share	numeric
if_flag	boolean
db_flag	boolean
feature_means	jsonb
reasons	jsonb
first_anomaly	timestamptz

mv_anomaly_score hypertable

PF **run_id**	bigint
PK **system_id**	bigint
PK **ts**	timestamptz
split	text
regime	text
anomaly_score	float8
score_smooth	float8
model_flag	boolean
is_anomaly	boolean
rule_flags	text[]
model_scores	jsonb
models_flagging	smallint
top_contributors	jsonb

arb_study_day

PF **study_id**	bigint
PK **kpi_date**	date
FK run_id	bigint
solver_status	text
solve_seconds	numeric(10,3)
expected_profit_usd	numeric(12,2)
gross_usd	numeric(12,4)
degradation_usd	numeric(12,4)
grid_usd	numeric(12,4)
net_usd	numeric(12,4)
oracle_net_usd	numeric(12,4)
pack_net_usd	numeric(12,4)
discharge_mwh	numeric(12,6)
charge_mwh	numeric(12,6)
cycles	numeric(8,4)
soc_start_pct	numeric(6,2)
soc_end_pct	numeric(6,2)
checks	jsonb

arb_backtest

PF **study_id**	bigint
PK **kind**	text
PK **interval_start**	timestamptz
FK run_id	bigint
p_kw	numeric(10,2)
c_as_mw	numeric(8,3)
soc_pct	numeric(6,2)
rt_price_mwh	numeric(10,2)
fc_price_mwh	numeric(10,2)
energy_mwh	numeric(12,6)
gross_usd	numeric(12,4)
degradation_usd	numeric(12,4)
grid_usd	numeric(12,4)
as_usd	numeric(12,4)
net_usd	numeric(12,4)

battery_itops

Activity log written by the observability framework

1 tables · 0 hypertables · 0 foreign keys · 0 views (see index) · dashed boxes are tables in other schemas

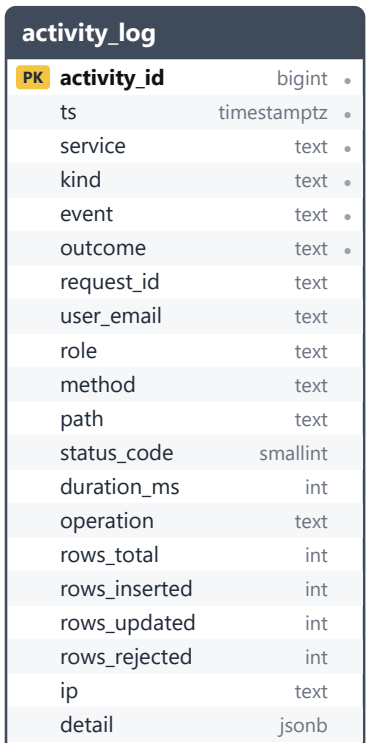


Table index (1/4)

Schema	Table	Cols	Primary key	References	Comment
battery_master	**aging_snapshot**	15	system_id, snapshot_date	battery_system	Monthly aging features + RUL labels (rul_days/rul_cycles); is_censored for survival models.
battery_master	**alarm_catalog**	12	alarm_catalog_id	alarm_severity	Alarm definitions per source system (BMS/PCS/HVAC/EMS); key (source, alarm_code)
battery_master	**alarm_severity**	2	severity_level		
battery_master	**as_award**	8	award_id	as_product, battery_system	Awarded AS commitments; each reserves power/energy headroom the optimizer must respect.
battery_master	**as_price** hypertable	5	product_id, delivery_start	as_product	Ancillary-service capacity/performance clearing prices per block.
battery_master	**as_product**	10	product_id	market	Ancillary-service products the asset can bid (reg, reserves, capacity).
battery_master	**asset_hierarchy**	7	asset_id	asset_hierarchy	
battery_master	**battery_chemistry**	5	chemistry_code		
battery_master	**battery_container**	10	container_id	battery_system	
battery_master	**battery_module**	12	module_id	battery_pack	
battery_master	**battery_opt_param**	15	param_id	battery_system	Per-system optimizer parameters: usable energy, SOC band, power/ramp limits, RTE, cycle cap, degradation cost.
battery_master	**battery_pack**	8	pack_id	battery_container	
battery_master	**battery_provenance**	12	system_id	battery_system	Cell provenance, batch, and BOL baseline per system.
battery_master	**battery_system**	16	battery_system_id	battery_chemistry, site	
battery_master	**calendar_day**	6	calendar_date		Daily calendar with holiday/day-type flags.
battery_master	**capacity_test**	10	test_id	battery_system	Periodic RPT capacity/resistance checkpoints (the SOH ground truth).
battery_master	**contactor_state**	2	state_code		
battery_master	**current_sensor_cal**	5	system_id, calibrated_on	battery_system	
battery_master	**dam_price** hypertable	5	node_id, delivery_start	price_node	Day-ahead market cleared energy prices per delivery block.
battery_master	**decision_state**	10	run_id	optimization_run	State the optimizer saw at horizon start (SOC/SOH/temp/available power/cycle budget/AS reservation).
battery_master	**degradation_cost_curve**	4	system_id, dod_band_pct	battery_system	Marginal $/MWh cycling cost by depth-of-discharge band; deeper cycles cost more (from RUL aging economics).
battery_master	**demand_tariff**	9	tariff_id	site	Demand-charge definition; sets what "peak" means economically.
battery_master	**dispatch_outcome** hypertable	11	system_id, interval_start	optimization_run, battery_system	Realized per-interval P&L: scheduled vs actual power, revenue net of cycling + grid cost (training label / scoring).
battery_master	**dispatch_schedule** hypertable	7	run_id, interval_start	optimization_run	Optimizer-produced setpoint schedule per interval (the decision to execute).
battery_master	**efficiency_curve**	5	system_id, direction, power_fraction, soc_band	battery_system	One-way efficiency lookup vs power fraction and SOC band (RTE is charge*discharge).
battery_master	**ems_controller**	8	ems_id	battery_system	
battery_master	**ems_dispatch_mode**	2	mode_code		
battery_master	**ems_grid_service**	2	service_code		
battery_master	**ems_tariff_period**	14	tariff_period_id	ems_controller	
battery_master	**eol_criteria**	4	chemistry		
battery_master	**fault_alarm_link**	3	episode_id, alarm_id	fault_episode	Maps raw alarm events to the confirmed fault episode they belong to (alarm grouping / correlation).
battery_master	**fault_catalog**	11	fault_code	alarm_severity	Labelled fault classes (the diagnosis/classification target set) — failure modes, not alarm codes.
battery_master	**fault_episode**	18	episode_id	fault_catalog, alarm_severity, battery_system	Confirmed, labelled fault instances (the supervised training labels); resolved_ts NULL = ongoing/censored.
battery_master	**fault_prediction**	13	prediction_id	fault_catalog, fault_episode, fault_catalog, battery_system	Classifier predictions vs confirmed labels — the basis for accuracy / precision-recall / confusion matrix.
battery_master	**fault_signature**	10	signature_id	fault_catalog	Symptom->fault diagnostic signatures: the a-priori feature/rule matrix for classification & root-cause ranking.
battery_master	**generation_asset**	6	gen_id	site	
battery_master	**grid_charge**	8	charge_id	site	Cost-stack components on energy moved (wheeling, transmission, demand charge, taxes).
battery_master	**hvac_cooling_mode**	2	mode_code		
battery_master	**hvac_unit**	9	hvac_id	battery_system	
battery_master	**interconnection_limit**	6	system_id, valid_from	battery_system	Point-of-interconnection export/import caps and charging source rule.
battery_master	**lifecycle_event**	8	event_id	battery_system	
battery_master	**maintenance_action**	15	action_id	fault_episode, battery_system	Repair/work-order record per episode: component, parts, cost, MTTR, whether the diagnosis and the fix held.
battery_master	**market**	9	market_id		Market / exchange a node settles in: currency, block lengths, gate closure.
battery_master	**metering_point**	6	metering_point_id	site	Revenue/feeder metering points where demand is measured.
battery_master	**ocv_curve**	5	chemistry, direction, temp_c, soc_pct		Per-cell OCV vs SOC by direction (hysteresis) and temperature.
battery_master	**operating_mode**	2	mode_code		
battery_master	**optimization_run**	12	run_id	battery_ml.arb_study, battery_system	One optimizer run: horizon, method, objective, solver status, expected profit.
battery_master	**pcs**	26	pcs_id	battery_system	A power conversion (inverter) unit within a BESS system.

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Schema	Table	Cols	Primary key	References	Comment
battery_master	**pcs_alarm_type**	4	alarm_type_code	alarm_severity	Catalog of PCS alarm/fault codes.
battery_master	**pcs_breaker_state**	2	state_code		
battery_master	**pcs_command_type**	2	command_code		
battery_master	**pcs_grid_service**	2	service_code		Grid-support functions the PCS can perform.
battery_master	**pcs_mode**	2	mode_code		PCS operating states (control modes).
battery_master	**pcs_ride_through_type**	2	rt_code		
battery_master	**price_forecast** hypertable	8	node_id, issued_at, target_ts	price_node	Probabilistic price forecast (quantiles) per issue vintage; horizon widens the band.
battery_master	**price_node**	6	node_id	market, site	Settlement node / bidding zone the plant is priced at.
battery_master	**price_scenario** hypertable	6	node_id, issued_at, scenario_id, target_ts	price_node	Sampled price paths + weights for stochastic/robust dispatch optimization.
battery_master	**rt_price** hypertable	4	node_id, interval_start	price_node	Real-time / imbalance settlement prices (what intraday deviations settle at).
battery_master	**site**	15	site_id		
battery_master	**soc_reference**	5	system_id, ts	battery_system	Independent ground-truth SOC (recalibration / lab anchors) — the label.
battery_master	**tenant**	16	tenant_id		
battery_master	**tenant_contact**	9	contact_id	tenant	
battery_master	**tenant_site**	5	tenant_id, site_id	site, tenant	
battery_master	**transformer**	6	transformer_id	site	
battery_master	**weather_station**	6	station_id	site	
battery_users	**access_request**	15	request_id	role, battery_master.tenant, app_user	
battery_users	**app_user**	16	id	role, battery_master.tenant	
battery_users	**auth_audit**	9	audit_id		
battery_users	**privilege**	5	privilege_id		
battery_users	**refresh_token**	11	token_id	app_user	
battery_users	**role**	5	role_id		
battery_users	**role_privilege**	2	role_id, privilege_id	privilege, role	
battery_raw	**bak_20261015_aging_snapshot**	15			
battery_raw	**bak_20261015_battery_provenance**	12			
battery_raw	**bak_20261015_bess_system**	2			
battery_raw	**bak_20261015_capacity_test**	10			
battery_raw	**bak_20261015_lifecycle_event**	8			
battery_raw	**bak_20261015_rul_backtest**	13			
battery_raw	**bak_20261015_rul_forecast**	15			
battery_raw	**bak_20261015_rul_run**	18			
battery_raw	**bak_20261015_rul_score**	13			
battery_raw	**batchlog**	26	batch_id		Load reports of the realtime snapshot batch: one row per input file processed
battery_raw	**soc_estimation_data**	17	stg_id		Staging copy of soc_estimation_data CSV files, as is; used to UPDATE system_telemetry SOC-estimation columns.
battery_raw	**soc_stg**	9	stg_id		Staging copy of soc_reference CSV files, as is (no mapping / validation); reloading a file replaces its rows.
battery_telemetry	**alarm_event** hypertable	18	time, alarm_id	battery_master.alarm_catalog, battery_master.battery_system, battery_master.alarm_severity	Alarm occurrences (raise -> clear); key (alarm_id, time); time = start time
battery_telemetry	**calendar_day**	4	calendar_date		
battery_telemetry	**demand_telemetry** hypertable	6	metering_point_id, ts	battery_master.metering_point	Feeder demand, behind-meter generation and net load (TimescaleDB hypertable).
battery_telemetry	**dispatch_event** hypertable	24	time, dispatch_id	battery_master.ems_dispatch_mode, battery_master.ems_controller, battery_master.ems_grid_service, battery_master.battery_system	EMS dispatch instructions; key (dispatch_id, time); time = effective_from
battery_telemetry	**ems_telemetry** hypertable	26	ems_id, ts	battery_master.ems_dispatch_mode, battery_master.ems_controller, battery_master.ems_grid_service	Plant/EMS telemetry (TimescaleDB hypertable): POI metering, dispatch, SOC, market — sampled every 5 min.
battery_telemetry	**hvac_telemetry** hypertable	26	hvac_id, ts	battery_master.hvac_cooling_mode, battery_master.hvac_unit	Container HVAC/TMS telemetry (TimescaleDB hypertable), sampled every 5 min.

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Schema	Table	Cols	Primary key	References	Comment
battery_telemetry	**weather_station**	2	station_id		
battery_ml	**anomaly_module_result**	18	run_id, module_id	anomaly_run	
battery_ml	**anomaly_point**	7	run_id, module_id, bucket	anomaly_run	
battery_ml	**anomaly_run**	31	run_id	battery_master.battery_system, anomaly_run	
battery_ml	**arb_backtest**	15	study_id, kind, interval_start	battery_master.optimization_run, arb_study	
battery_ml	**arb_study**	16	study_id	battery_master.price_node, battery_master.battery_system	
battery_ml	**arb_study_day**	18	study_id, kpi_date	battery_master.optimization_run, arb_study	
battery_ml	**aux_forecast_daily**	8	run_id, ems_id, day, method	aux_forecast_run	
battery_ml	**aux_forecast_point**	9	run_id, ems_id, origin_ts, horizon_min, method	aux_forecast_run	
battery_ml	**aux_forecast_run**	21	run_id		
battery_ml	**aux_forecast_score**	10	run_id, ems_id, horizon_bucket, method	aux_forecast_run	
battery_ml	**bms_kpi**	11	system_id, kpi_date, kpi_code	bms_kpi_run, battery_master.battery_system	Daily BMS KPIs per battery system, computed from system_telemetry by the definitions in kpi_config/bms_kpi.yaml.
battery_ml	**bms_kpi_run**	18	run_id		One row per BMS KPI calculation (API button or CLI).
battery_ml	**efficiency_cycle**	31	run_id, cycle_no	efficiency_run	
battery_ml	**efficiency_run**	29	run_id		
battery_ml	**efficiency_score**	12	run_id, model, dataset	efficiency_run	
battery_ml	**ems_kpi**	11	ems_id, kpi_date, kpi_code	battery_master.ems_controller, ems_kpi_run	Daily EMS KPIs per EMS controller, computed from ems_telemetry by the definitions in kpi_config/ems_kpi.yaml.
battery_ml	**ems_kpi_run**	18	run_id		One row per EMS KPI calculation (API button or CLI).
battery_ml	**fault_diag_candidate**	19	candidate_id	battery_master.fault_catalog, battery_master.fault_episode, battery_master.fault_catalog, battery_master.fault_prediction, fault_diag_run	
battery_ml	**fault_diag_example**	14	run_id, example_key	fault_diag_run	
battery_ml	**fault_diag_run**	16	run_id		
battery_ml	**fault_diag_scan_point**	8	run_id, system_id, ts	fault_diag_run	
battery_ml	**heatload_forecast_point** hypertable	12	run_id, system_id, issued_at, horizon_min, valid_at	heatload_run	
battery_ml	**heatload_ledger** hypertable	25	run_id, system_id, ts	heatload_run	
battery_ml	**heatload_run**	14	run_id		
battery_ml	**hvac_cop_point** hypertable	27	run_id, hvac_id, ts	hvac_cop_run	
battery_ml	**hvac_cop_run**	14	run_id		
battery_ml	**loss_daily**	42	run_id, system_id, day	loss_run	
battery_ml	**loss_ledger** hypertable	25	run_id, system_id, ts	loss_run	
battery_ml	**loss_run**	14	run_id		
battery_ml	**model_run**	62	run_id	model_run	SOH model training runs (XGBoost / LightGBM) with their scores
battery_ml	**model_run_curve**	6	run_id, iteration	model_run	Train/validation error per boosting round of a model run
battery_ml	**mv_anomaly_run**	21	run_id	mv_anomaly_run, battery_master.battery_system	Multivariate anomaly detection runs: model builds (train), scorings of other windows (score), live micro-batch target (live).
battery_ml	**mv_anomaly_score** hypertable	13	run_id, system_id, ts	mv_anomaly_run	Per-interval multivariate anomaly score, flag and top contributing features (sibling of system_telemetry).
battery_ml	**netload_forecast_peak**	14	run_id, mp_id, origin_ts, target_date, model	netload_forecast_run	
battery_ml	**netload_forecast_point**	10	run_id, mp_id, origin_ts, step, model	netload_forecast_run	
battery_ml	**netload_forecast_run**	29	run_id		
battery_ml	**netload_forecast_score**	19	run_id, model	netload_forecast_run	
battery_ml	**pcs_kpi**	11	pcs_id, kpi_date, kpi_code	battery_master.pcs, pcs_kpi_run	Daily PCS KPIs per PCS unit, computed from pcs_telemetry by the definitions in kpi_config/pcs_kpi.yaml.
battery_ml	**pcs_kpi_run**	18	run_id		One row per PCS KPI calculation (API button or CLI).
battery_ml	**price_forecast_point**	12	run_id, ems_id, origin_ts, horizon_h	price_forecast_run	

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Schema	Table	Cols	Primary key	References	Comment
battery_ml	**soc_estimation_run**	24	run_id		
battery_ml	**soc_estimation_score**	12	run_id, method, scope	soc_estimation_run	
battery_ml	**soc_estimation_state**	5	run_id, system_id	soc_estimation_run	
battery_ml	**soc_forecast_point**	10	run_id, ems_id, origin_ts, step, method	soc_forecast_run	
battery_ml	**soc_forecast_run**	23	run_id		
battery_ml	**soc_forecast_score**	14	run_id, method	soc_forecast_run	
battery_ml	**thermal_forecast_point**	11	run_id, target, system_id, origin_ts, horizon_min, method	thermal_forecast_run	
battery_ml	**thermal_forecast_run**	22	run_id		
battery_ml	**thermal_forecast_score**	15	run_id, target, horizon_min, method	thermal_forecast_run	
battery_itops	**activity_log**	20	activity_id		

View index

Views and TimescaleDB continuous aggregates, with the relations each one reads from.

Schema	View	Type	Cols	Reads from
battery_master	**dispatch_pnl_daily**	continuous aggregate	8	(continuous aggregate storage)
battery_master	**v_arbitrage_features**	view	19	battery_opt_param, battery_system, dam_price, interconnection_limit, price_forecast, price_node, rt_price
battery_master	**v_dispatch_pnl**	view	9	dispatch_outcome
battery_master	**v_fault_scoring**	view	6	fault_prediction
battery_master	**v_fault_signature_profile**	view	6	fault_catalog, fault_signature
battery_master	**v_fault_training**	view	28	battery_system, fault_alarm_link, fault_catalog, fault_episode, maintenance_action
battery_master	**v_rul_training**	view	20	aging_snapshot, battery_provenance, battery_system
battery_master	**v_soc_features**	view	19	soc_reference, battery_telemetry.system_telemetry
battery_master	**v_thermal_assets**	view	4	battery_system, ems_controller, hvac_unit, pcs
battery_users	**v_role_permissions**	view	6	privilege, role, role_privilege
battery_telemetry	**demand_daily**	continuous aggregate	10	(continuous aggregate storage)
battery_telemetry	**ems_aux_15m**	continuous aggregate	5	(continuous aggregate storage)
battery_telemetry	**ems_telemetry_daily**	continuous aggregate	19	(continuous aggregate storage)
battery_telemetry	**hvac_telemetry_daily**	continuous aggregate	19	(continuous aggregate storage)
battery_telemetry	**hvac_thermal_15m**	continuous aggregate	8	(continuous aggregate storage)
battery_telemetry	**module_telemetry_daily**	continuous aggregate	11	(continuous aggregate storage)
battery_telemetry	**pcs_telemetry_daily**	continuous aggregate	16	(continuous aggregate storage)
battery_telemetry	**pcs_thermal_15m**	continuous aggregate	8	(continuous aggregate storage)
battery_telemetry	**system_telemetry_daily**	continuous aggregate	18	(continuous aggregate storage)
battery_telemetry	**system_thermal_15m**	continuous aggregate	8	(continuous aggregate storage)
battery_telemetry	**v_aux_features**	view	32	battery_master.battery_system, battery_master.ems_controller, battery_master.v_thermal_assets, battery_master.weather_station, ems_telemetry, hvac_telemetry, pcs_telemetry, system_telemetry, v_aux_rolling
battery_telemetry	**v_aux_rolling**	view	6	battery_master.ems_controller, battery_master.v_thermal_assets, ems_aux_15m, hvac_thermal_15m
battery_telemetry	**v_ems_latest**	view	13	battery_master.ems_controller, ems_telemetry
battery_telemetry	**v_hvac_latest**	view	12	battery_master.hvac_unit, hvac_telemetry
battery_telemetry	**v_module_latest**	view	8	battery_master.battery_module, module_telemetry
battery_telemetry	**v_netload_features**	view	17	battery_master.calendar_day, demand_telemetry, weather_actual
battery_telemetry	**v_soh_daily_features**	view	29	battery_master.battery_system, system_telemetry
battery_telemetry	**v_soh_features_5min**	view	21	system_telemetry
battery_telemetry	**v_thermal_features**	view	40	battery_master.v_thermal_assets, ems_telemetry, hvac_telemetry, pcs_telemetry, system_telemetry, v_thermal_rolling_1h
battery_telemetry	**v_thermal_rolling_1h**	view	10	battery_master.v_thermal_assets, hvac_thermal_15m, pcs_thermal_15m, system_thermal_15m
battery_ml	**v_heatload_daily**	view	20	heatload_ledger
battery_ml	**v_hvac_cop_daily**	view	12	hvac_cop_point
battery_ml	**v_mv_anomaly_alarm**	view	17	battery_master.alarm_catalog, mv_anomaly_score, battery_telemetry.alarm_event